🚊 PASHUPATASTRA

AI-Assisted Railway Maintenance Scheduling & Disruption Recovery

Document: Phase 1 Domain Review & Synthetic Railway Data Specification
Workstream Owner: Darshini (Domain Modeling & Synthetic Data)
Branch: `feature/domain-data`

Event: Smart India Hackathon 2026
Integration Owner: Aryan
Status: Complete & Verified (Ready for Domain Freeze)

1. Executive Summary & Purpose

This document formalizes the Indian Railways operational domain modeling, shared data schemas, and reproducible synthetic corridor scenarios for **Phase 1** of Project Pashupatastra. All downstream modules (CP-SAT Optimizer by Tyagi, ML Scorer by Ayush, Disruption Simulation by Tirth, Dashboard UI by Archit, and FastAPI Backend by Aryan) consume the models and fixtures documented herein.

2. Domain Review & Decisions (Domain Freeze Checkpoint)

2.1 Track Modeling

Decision: Retain flat string `track_id` (e.g. `"UP-1"`, `"DOWN-1"`) as the primary key in CP-SAT constraints for solver performance, while attaching structured corridor metadata (section name, mileage, direction, speed limit) in the data layer for UI filtering.

2.2 Maintenance Work Types

Standardized 6 concrete Indian Railways maintenance categories mapped to physical operations and machinery:

Work Category	Operational Description	Typical Duration	Required Machinery / Gang	Priority Range
`TRACK_RENEWAL`	Complete Track Renewal (CTR), Sleeper & Rail renewal	180 – 240 min	`BCM_MACHINE_01` (Ballast Cleaner)	**0.85 – 0.95**
`BALLAST_TAMPING`	Track packing, alignment & geometry correction	90 – 150 min	`CSM_TAMPER_01` (Continuous Tamper)	**0.70 – 0.85**
`OHE_MAINTENANCE`	Overhead Electrification inspection & isolations	75 – 120 min	`TOWER_WAGON_01` (Inspection Unit)	**0.65 – 0.80**
`SIGNALLING_INTERLOCKING`	Point machine overhaul, track circuits, axle counters	60 – 120 min	`S_T_GANG_01` (Signal & Telecom Gang)	**0.60 – 0.75**
`ROUTINE_INSPECTION`	USFD ultrasonic rail flaw detection & foot patrol	45 – 90 min	*Manual Trolley Gang*	**0.35 – 0.50**
`EMERGENCY_REPAIR`	Rail fracture, weld failure, snapped OHE wire	90 – 150 min	*Emergency Flying Squad*	**0.95 – 0.99**

2.3 Resource & Equipment Constraints

Decision: Avoid multi-resource bin-packing that bloats solver complexity. Shared heavy machines (BCM, CSM Tamper, Tower Wagon) are constrained using `mutual_exclusion_group` intervals, preventing simultaneous overlapping use across different tracks.

2.4 Train Timetable & Operational Possession Windows

Train movements are modeled via discrete available **Possession Windows** per track over a 24-hour horizon (1440 min):

Window ID	Track	Window Type	Clock Time	Minute Interval	Operational Purpose
`POS-UP1-NIGHT`	`UP-1`	NIGHT_TRAFFIC_BLOCK	00:30 – 05:00	**0030 – 0300** (270 min)	Primary heavy renewals & tamping
`POS-UP1-MIDDAY`	`UP-1`	MIDDAY_MAINTENANCE	11:30 – 14:30	**0690 – 0870** (180 min)	Inspections & OHE maintenance
`POS-UP1-EVENING`	`UP-1`	EVENING_OFF_PEAK	21:00 – 23:30	**1260 – 1410** (150 min)	Signalling & minor adjustments
`POS-DN1-NIGHT`	`DOWN-1`	NIGHT_TRAFFIC_BLOCK	00:30 – 05:00	**0030 – 0300** (270 min)	Primary heavy renewals & tamping
`POS-DN1-MIDDAY`	`DOWN-1`	MIDDAY_MAINTENANCE	11:30 – 14:30	**0690 – 0870** (180 min)	Inspections & OHE maintenance
`POS-DN1-EVENING`	`DOWN-1`	EVENING_OFF_PEAK	21:00 – 23:30	**1260 – 1410** (150 min)	Signalling & minor adjustments

3. Synthetic Scenario Datasets

3.1 Scenario A: Baseline Mainline Corridor (`corridor_a_blocks.json`)

Corridor: `CORRIDOR_A` (UP-1, DOWN-1) | **Candidates:** 12 Blocks | **Horizon:** 1440 min | **Headway:** 15 min

Block ID	Track	Asset ID	Work Category	Dur	Pri	Risk	Allowed Window	Machine Group	Dependency
BLK-001	DOWN-1	AST-DN-OHE-013	OHE_MAINTENANCE	120 m	0.79	0.63	11:00 – 15:30 (0660-0930)	TOWER_WAGON_01	—
BLK-002	DOWN-1	AST-DN-RAI-015	BALLAST_TAMPING	105 m	0.50	0.50	00:30 – 06:00 (0030-0360)	CSM_TAMPER_01	—
BLK-003	DOWN-1	AST-DN-RAI-014	TRACK_RENEWAL	210 m	0.59	0.66	00:30 – 06:00 (0030-0360)	BCM_MACHINE_01	—
BLK-004	DOWN-1	AST-DN-RAI-009	BALLAST_TAMPING	150 m	0.59	0.59	04:00 – 07:30 (0240-0450)	CSM_TAMPER_01	BLK-003 🔗
BLK-005	UP-1	AST-UP-OHE-004	OHE_MAINTENANCE	75 m	0.56	0.52	00:30 – 23:30 (0030-1410)	TOWER_WAGON_01	—
BLK-006	DOWN-1	AST-DN-OHE-016	OHE_MAINTENANCE	120 m	0.54	0.49	00:30 – 23:30 (0030-1410)	TOWER_WAGON_01	—
BLK-007	DOWN-1	AST-DN-OHE-011	OHE_MAINTENANCE	120 m	0.79	0.53	11:00 – 15:30 (0660-0930)	TOWER_WAGON_01	—
BLK-008	UP-1	AST-UP-RAI-001	TRACK_RENEWAL	210 m	0.78	0.77	00:30 – 06:00 (0030-0360)	BCM_MACHINE_01	—
BLK-009	UP-1	AST-UP-TUR-007	SIGNALLING_INTERLOCK	105 m	0.42	0.44	00:30 – 23:30 (0030-1410)	S_T_GANG_01	—
BLK-010	UP-1	AST-UP-RAI-008	ROUTINE_INSPECTION	45 m	0.42	0.46	11:00 – 15:30 (0660-0930)	—	—
BLK-011	UP-1	AST-UP-SIG-002	SIGNALLING_INTERLOCK	75 m	0.65	0.59	00:30 – 23:30 (0030-1410)	S_T_GANG_01	—

Block ID	Track	Asset ID	Work Category	Dur	Pri	Risk	Allowed Window	Machine Group	Dependency
`BLK-012`	`UP-1`	`AST-UP-TUR-006`	SIGNALLING_INTERLOCK	75 m	0.57	0.50	00:30 – 23:30 (0030-1410)	`S_T_GANG_01`	—

3.2 Scenario B: High Density 4-Track Trunk Corridor (`corridor_b_dense.json`)

Corridor: `CORRIDOR_B_DENSE` | **Tracks:** `UP-1`, `UP-2`, `DOWN-1`, `DOWN-2` | **Candidates:** 24 Blocks
Simulates heavy quadruple trunk routes (e.g. New Delhi – Kanpur). Contains 4 competing track machine groups with tight possession windows to test CP-SAT optimization and solver trade-offs under saturation.

3.3 Scenario C: Disruption & Recovery Testbed (`corridor_c_disrupted.json`)

Corridor: `CORRIDOR_C_DISRUPTED` | **Tracks:** `UP-1`, `DOWN-1` | **Candidates:** 14 Blocks
Pre-configured with **2 COMMITTED blocks** (locked on `DOWN-1` from 01:00 to 03:15). Ready for Tirth's disruption simulator to inject `TRACK_UNAVAILABLE` on `UP-1` from 02:00 to 05:00 and test CP-SAT recovery.

4. Code & Artifact Checklist

✓ **DOMAIN_REVIEW.md:** `C:\Users\Hp\.gemini\antigravity\scratch\pashupatastra\DOMAIN_REVIEW.md`

✓ **Data Generator:** `C:\Users\Hp\.gemini\antigravity\scratch\pashupatastra\backend\app\data\generator.py`

✓ **Domain Models:** `C:\Users\Hp\.gemini\antigravity\scratch\pashupatastra\backend\app\data\models.py`

✓ **JSON Fixtures:** `backend/app/data/fixtures/` (`corridor_a_blocks.json`, `corridor_b_dense.json`, `corridor_c_disrupted.json`)

✓ **Unit Test Suite:** `backend/tests/test_generator.py` (**10/10 tests passed** in 0.049s)

✓ **CLI Runner:** `generate_data.py` (Run `python generate_data.py --scenario A`)

> **Repository Sync:** All files are mirrored at `D:\priyadarshini projects\pashupatastra\` and `C:\Users\Hp\.gemini\antigravity\scratch\pashupatastra\` .